Training and validation curves

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Freeze the checkpoint. Evaluate both splits. Restore the selected state.

Read comparable learning curves and choose checkpoints with a declared validation rule.

Keep one habit: Complete saved states and paired losses → full curves/zoom → pictured selection change → tie/metric/status probes → independent new split targets.

MAIN LESSON · SECTION 1

Which saved state does the declared rule actually choose?

> Evaluate both fixed splits at the same checkpoint without updating during evaluation.

Learning goals

I can compute comparable **training** and validation losses at saved **checkpoint**s.

I can choose among observed **checkpoint**s using a declared validation rule.

I can distinguish development **evidence** from final test **evidence**.

Retrieve controlled training comparisons

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Why does the lowest **training** loss in a learning-rate experiment not establish the best performance on new examples?

> **My starting explanation**

End goal

By the end, I can explain **training** and **validation curve**s using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Compute comparable **training** and validation losses at saved **checkpoint**s

Choose among observed **checkpoint**s using a declared validation rule

Distinguish development **evidence** from final test **evidence**

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Keep the model state fixed while evaluating each split

> The constructed target rules differ;the curves demonstrate selection,not a diagnosed real population.

Complete training history and distinct validation pairs

Keep the restricted network h = ReLU(x), y = vh, with only v trained. **Training** pairs remain (1, 2) and (2, 4). Start v = 0 and make four full-data updates at rate 0.1. Save states v = 0, 1, 1.5, 1.75 and 1.875.

At each saved state, evaluate both splits without changing v during evaluation.

For this deliberately constructed demonstration, validation pairs are (3, 4.5) and (4, 6). Their targets follow a different slope from the **training** pairs.

This makes the two trends disagree visibly; it is not **evidence** that these few rows represent a real population. Validation labels do not enter the gradient.

All five same-state split evaluations

Updates	v	Training MSE	Validation MSE
0	0	10	28.125
1	1	2.5	3.125
2	1.5	0.625	0
3	1.75	0.15625	0.78125
4	1.875	0.0390625	1.7578125

Mean-loss definitions and known split differences

Both columns use ordinary MSE in the same target units, evaluated on their own fixed rows. **Training** MSE is 2.5(v - 2) squared; validation MSE is 12.5(v - 1.5) squared.

Different **input** values and target rules affect the loss levels. A gap alone does not identify its cause.

Name given and missing facts

List the given facts and one fact that is not supplied.

Trace both split losses back to paired predictions

> Compare the same metric and units at one saved model state.

Complete checkpoint3calculation

For **checkpoint** 3, v = 1.75. **Training prediction**s are [1.75, 3.5], residuals [-0.25, -0.5], and MSE 0.15625. Validation **prediction**s are [5.25, 7], residuals [0.75, 1], and MSE 0.78125.

All four **prediction**s must use the same saved value.

Complete checkpoint 3(v1.75)paired calculation · Input x · Target · Prediction vReLU(x) · Residual prediction–target · Squared error

> **Complete checkpoint 3(v1.75)paired calculation:** Training A
>
> **Input x:** 1
>
> **Target:** 2
>
> **Prediction vReLU(x):** 1.75
>
> **Residual prediction–target:** -0.25
>
> **Squared error:** 0.0625

> **Complete checkpoint 3(v1.75)paired calculation:** Training B
>
> **Input x:** 2
>
> **Target:** 4
>
> **Prediction vReLU(x):** 3.5
>
> **Residual prediction–target:** -0.5
>
> **Squared error:** 0.25

> **Complete checkpoint 3(v1.75)paired calculation:** Validation C
>
> **Input x:** 3
>
> **Target:** 4.5
>
> **Prediction vReLU(x):** 5.25

Residual prediction–target: 0.75

Squared error: 0.5625

Complete checkpoint 3(v1.75)paired calculation: Validation D

Input x: 4

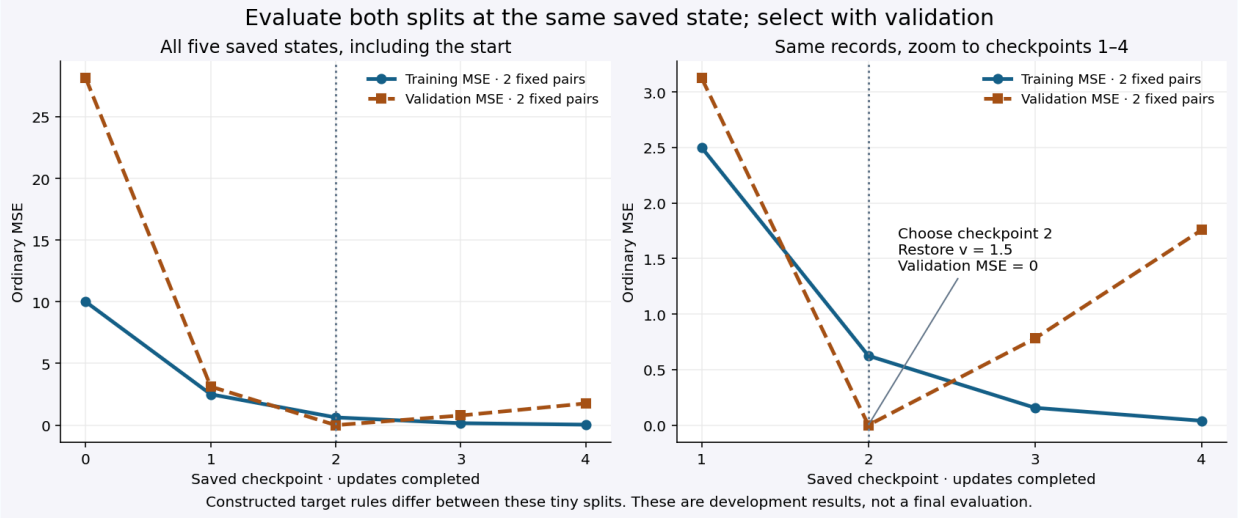
Target: 6

Prediction vReLU(x): 7

Residual prediction–target: 1

Squared error: 1

Complete five-checkpoint curves and zoom



Checkpoint-evaluation and restoration contract

The current program evaluates the five supplied saved vvalues; it does not run training or change vusing validation gradients. The training sequence can be reproduced from the earlier complete four-update experiment. Each saved state is held fixed while both training and

validation MSEs are computed. Choose lowest observed validation MSE, then earliest index on an exact tie.

Restore the selected saved parameter v1.5, not the last loop variable v1.875. Zero loss on two constructed validation cases is limited model-selection evidence, not a final evaluation.

The code evaluates supplied **checkpoint**s; it does not train on validation data. It assumes finite numeric pairs and selects the lowest observed validation MSE, breaking exact ties in favour of the earlier **checkpoint**.

The rule is chosen before inspecting the values.

Runnable paired evaluator and earliest-tie selection

```
def mse(v, rows):
    if not rows:
        raise ValueError("Need evaluation rows")
    return sum((v * max(0.0, x) - target) ** 2
               for x, target in rows) / len(rows)

train = [(1.0, 2.0), (2.0, 4.0)]
valid = [(3.0, 4.5), (4.0, 6.0)]
checkpoints = [0.0, 1.0, 1.5, 1.75, 1.875]
validation_losses = []
for step, v in enumerate(checkpoints):
    validation_losses.append(mse(v, valid))
    print(step, mse(v, train), mse(v, valid))
best = min(range(len(checkpoints)),
           key=lambda i: (validation_losses[i], i))
print("Selected checkpoint", best)  # 2
```

Connect each plotted value to its source pairs

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Choose by the stated criterion,including its tie rule

> A smaller training loss does not override a validation-based selection rule.

Current exact-tie selection task

- Observed validation MSE at **checkpoint**s 0 to 4 is [5, 2, 1, 1, 1.2]. Lowest loss wins with earlier exact ties. Which **checkpoint** is selected? What if **training** loss is lowest at **checkpoint** 4?

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured revised validation value

- Change: Epoch 3 validation loss is instead 1.5.

▼ **Check after predicting**

Prediction: Select epoch 3 under the unchanged criterion.

Reason: Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read each curve’s meaning before choosing a checkpoint

> Keep the supplied paper log separate from the fully reproducible numeric fixture.

Complete paper-log facts and separate scope

Plot **training** and validation loss for **checkpoint**s before opening final cases.

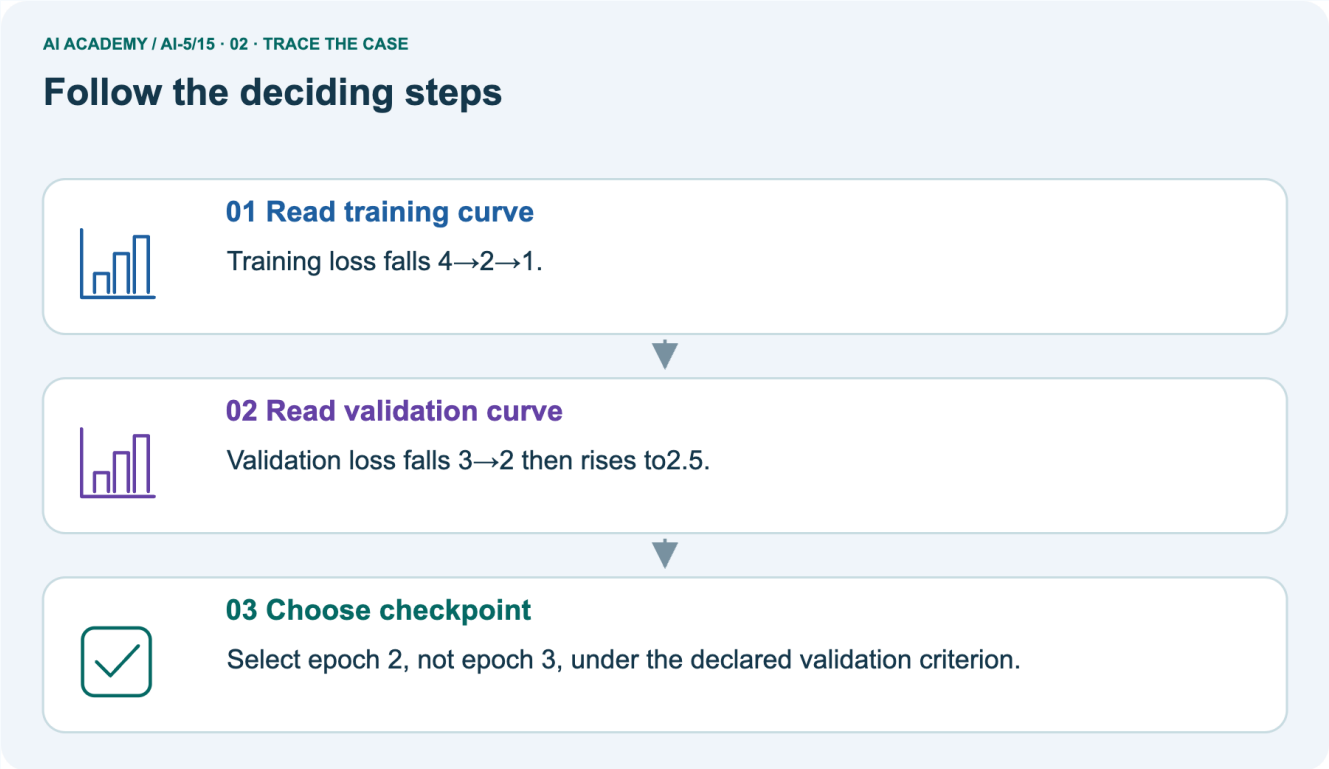
Supplied **checkpoint** losses: epoch 1 train 4, val 3; epoch 2 train 2, val 2; epoch 3 train 1, val 2.5. Selection: smallest validation loss.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate supplied three-epoch log:training 4/2/1, validation 3/2/2.5. Minimum validation chooses epoch 2, even though training is lowest at epoch 3. If only epoch 3 validation changes to1.5, the same rule instead chooses epoch 3. No training code or row-level predictions are supplied for that paper log.

The main executable five-checkpoint fixture includes all four numeric data pairs and chooses checkpoint 2, v1.5; its checkpoint count is not the pictured epoch numbering.

Trace training,validation and the selection rule



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read training curve	Training loss falls 4→2→1.
2. Read validation curve	Validation loss falls 3→2 then rises to2.5.
3. Choose checkpoint	Select epoch 2, not epoch 3, under the declared validation criterion.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

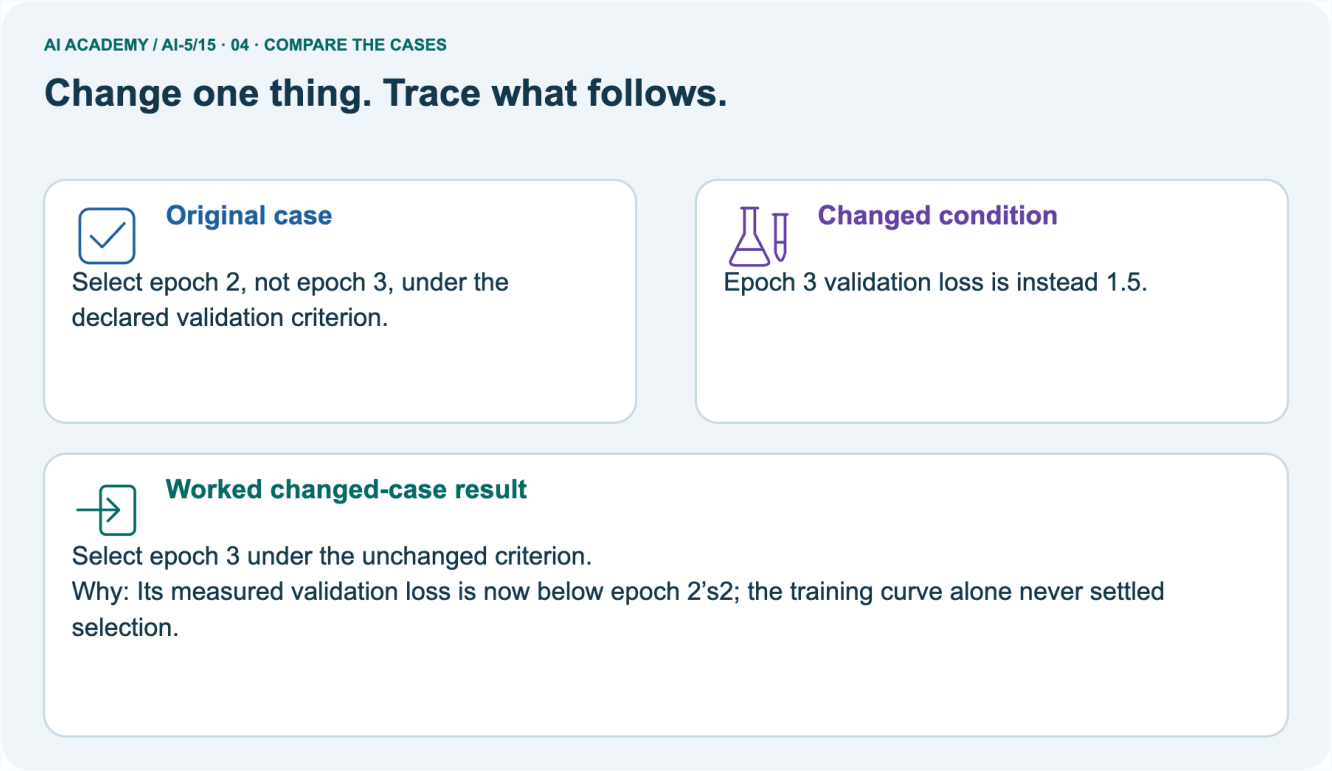
The **training curve** measures fit to examples used for learning. The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen.

Under the supplied **selection rule**, keep **checkpoint** two. A later final test must remain outside both learning and **checkpoint** selection.

Predict the one changed validation value

- Change one thing: Epoch 3 validation loss is instead 1.5. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain development status and uncertainty

- A single curve has sampling and run variability; validation-selected **checkpoint**s still require untouched final **evidence**.

- Read the labelled picture: Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try main checkpoint2to4loss changes

Later updates improve only one split

Series	2	3	4
Training solid line	0.625	0.15625	0.0390625
Validation dashed line	0	0.78125	1.7578125

This chart shows checkpoints 2 to 4 only, on one MSE scale from 0 to 2. Circles and a solid line mark training; squares and a dashed line mark validation. Lines connect measured checkpoints, not additional evaluations. The full table includes checkpoints 0 and 1.

Using the table, calculate the training and validation loss changes from checkpoint 2 to checkpoint 4. Which saved parameter does the selection rule choose?

▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Training and validation curves

The facts we will use

AI ACADEMY / AI-5/15

Checkpoint losses

	training	validation
epoch 1	4	3
epoch 2	2	2
epoch 3	1	2.5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/15

Training and validation curves

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read training curve

Training loss falls 4 -> 2 -> 1.



2. Read validation curve

Validation loss falls 3 -> 2 then rises to2.5.



3. Choose checkpoint

Select epoch 2, not epoch 3, under the declared validation criterion.

CHANGE ONE THING

Epoch 3 validation loss is instead 1.5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair incompatible metrics and unsupported diagnoses

> A gap alone does not isolate why the two splits behave differently.

Known target-rule difference and valid comparison requirements

Falling **training** loss with rising validation loss can be consistent with over**fitting**, but it is not a complete diagnosis. In this example the split-specific target rules are known to differ.

In another dataset, inspect representativeness, pre**process**ing, leakage, label quality and evaluation settings before attributing the pattern to model capacity alone.

Evaluate comparable quantities: do not compare a **training** sum with a validation mean, or a **training** objective containing a penalty with unpenalised validation MSE. Keep the row sets fixed across **checkpoint**s.

Do not mix losses recorded before each batch update with a full-split evaluation of the final **checkpoint**.

Convert training sum to a comparable mean

- A **training** report shows sum of squared errors 12 over six rows. Validation shows mean squared error 3 over two rows in the same target units. A learner says **training** error is four times larger. Repair the comparison.

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise checkpoint choice with less help

> Retain the criterion,all observed losses and restored parameter value.

Current task and response

- At v = 1, compute both validation **prediction**s, residuals and MSE on (3, 4.5) and (4, 6).

Using the table, calculate the training and validation loss changes from checkpoint 2 to checkpoint 4. Which saved parameter does the selection rule choose?

Explain your prediction to a partner before comparing results.

Supported W2record

Use workbook W2 for the connected example “**Training** and **validation curve**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Training** and **validation curve**s. Proposed explanation: “The last **checkpoint** is always best”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured curve interpretation

- Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve the independent new validation pairs

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Use the declared rule

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete same-state evaluation and selection audit

> All data,checkpoints,metric definitions and selection rules are supplied.

Current tie task and complete executable evaluator

Observed validation MSE at **checkpoint**s 0 to 4 is [5, 2, 1, 1, 1.2]. Lowest loss wins with earlier exact ties. Which **checkpoint** is selected? What if **training** loss is lowest at **checkpoint** 4?

```
def mse(v, rows):
    if not rows:
        raise ValueError("Need evaluation rows")
    return sum((v * max(0.0, x) - target) ** 2
               for x, target in rows) / len(rows)

train = [(1.0, 2.0), (2.0, 4.0)]
valid = [(3.0, 4.5), (4.0, 6.0)]
checkpoints = [0.0, 1.0, 1.5, 1.75, 1.875]
validation_losses = []
for step, v in enumerate(checkpoints):
    validation_losses.append(mse(v, valid))
    print(step, mse(v, train), mse(v, valid))
best = min(range(len(checkpoints)),
           key=lambda i: (validation_losses[i], i))
print("Selected checkpoint", best)  # 2
```

Complete normal,tie,missing,metric and evidence-status probes

Complete checkpoint-evaluation probe	Expected result / action
Normal main five states	Choose checkpoint 2, v1.5, validation 0, training.625
Exact-tie log[5, 2, 1, 1, 1.2]	Choose index 2 over3; training minimum at4 does not alter the rule
Evaluate checkpoint 3, v1.75	Training MSE.15625, validation.78125; all four predictions use v1.75
Missing/empty evaluation rows	Empty rows raise ValueError:Need evaluation rows; missing numeric references require review before calling
Training SSE 12on6 rows; validation MSE 3on2 rows	Comparable training MSE 2, validation MSE 3; do not compare 12 directly with 3
Mean loss plus penalty versus unpenalized validation MSE	Separate the penalty and compare the same predictive-loss definition
Choose a new checkpoint using final-test results	Those cases become selection evidence for the revision; reserve fresh final data
Zero selected validation loss	Exact fit on two constructed development rows; no universal or untouched-final claim

Predict,evaluate,restore and bound the result

1. Predict all five training/validation loss pairs before running or tracing the evaluator. Keep rows fixed and model values frozen during each evaluation. Record the chosen index and restore its saved vvalue explicitly. Trace the separate exact-tie list and sum-to-mean repair. The helper checks nonempty rows but assumes finite numeric pairs; it does not validate missing values.

   Record the known synthetic target-rule difference and investigate other possible causes in an actual project rather than calling every widening gap a complexity diagnosis. Final labels remain outside checkpoint selection.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: evaluate new split targets and candidate weights

> Calculate both columns before applying the stated validation rule.

Complete independent candidates and pairs

Use h = ReLU(x), y = vh, candidate v values 1, 1.5 and 2. **Training** pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE.

Explain the disagreement with selection by **training** loss.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Evaluate new checkpoints independently

Attempt this without hints

Use h = ReLU(x), y = vh, candidate v values 1, 1.5 and 2. **Training** pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE.

Explain the disagreement with selection by **training** loss.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the selected checkpoint with its selection history

> The project record should distinguish the saved final training state from the chosen model.

Project artifact and evidence table

Add a labelled **training** and **validation curve**s **evidence** sheet to Image-pattern Network Investigation. Show how you can compute comparable **training** and validation losses at saved **checkpoint**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a restored state and a bounded curve interpretation

> Explain which cases informed the choice and which evidence is still needed.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Compute comparable **training** and validation losses at saved **checkpoint**s

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Checkpoint	Saved model parameters and the information needed to identify a training state
Training curve	A record of a training measure at successive saved states
Validation curve	A record of a development measure on separate examples at those states
Selection rule	A declared procedure for choosing among candidate models or checkpoints

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- At v = 1, compute both validation **prediction**s, residuals and MSE on (3, 4.5) and (4, 6).
- A **training** report shows sum of squared errors 12 over six rows. Validation shows mean squared error 3 over two rows in the same target units. A learner says **training** error is four times larger. Repair the comparison.
- **Training** loss falls while validation loss rises. Give two checks before declaring model complexity the cause, and explain why repeatedly selecting with validation makes it development **evidence**.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent candidate weights and validation targets for the first individual attempt.

Current explanation and response

- Restore **checkpoint** 2, v = 1.5, if following the declared rule. Its zero validation MSE is a result on two constructed development examples. Choosing this **checkpoint** using those examples makes them part of model selection.

  After all choices are frozen, a separate untouched test set can provide final **evidence** within its own coverage limits.

- Next week adds a weight penalty and an early-stopping policy, keeping their roles distinct.

- **Training** loss falls while validation loss rises. Give two checks before declaring model complexity the cause, and explain why repeatedly selecting with validation makes it development **evidence**.

A team calls checkpoint 2 independently proven because its validation MSE is zero, then uses the test results to choose another checkpoint. Identify both reporting problems and propose a clean procedure.

My limited claim and a useful next check

▼ Extra practice · Revisit the complete trace

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Later updates improve only one split

Series	2	3	4
Training solid line	0.625	0.15625	0.0390625
Validation dashed line	0	0.78125	1.7578125

This chart shows checkpoints 2 to 4 only, on one MSE scale from 0 to 2. Circles and a solid line mark training; squares and a dashed line mark validation. Lines connect measured checkpoints, not additional evaluations. The full table includes checkpoints 0 and 1.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Plot **training** and validation loss for **checkpoint**s before opening final cases.

By the end, I can	How I will show it
Compute comparable training and validation losses at saved checkpoints	Explain the deciding step, show a trace or test, and state one limit.
Choose among observed checkpoints using a declared validation rule	Explain the deciding step, show a trace or test, and state one limit.
Distinguish development evidence from final test evidence	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Plot **training** and validation loss for **checkpoint**s before opening final cases.
- Which labelled step matches this detail? Select epoch 2, not epoch 3, under the declared validation criterion.
- 1. Read **training curve**
- 2. Choose **checkpoint**
- 3. Read **validation curve**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Choose **checkpoint** Select epoch 2, not epoch 3, under the declared validation criterion. The **training curve** measures fit to examples used for learning.

The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen. Under the supplied **selection rule**, keep **checkpoint** two.

A later final test must remain outside both learning and **checkpoint** selection.

Changed-case question and optional hint

- Changed-case practice: Epoch 3 validation loss is instead 1.5. Which conclusion is supported by the supplied facts?
- 1. Select epoch 3 under the unchanged criterion.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Select epoch 3 under the unchanged criterion. Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

Repair and retain the record

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small network (ai-5/12). Goal: Trace repeated forward loss gradient and update steps. Short practice: Which parameter changed, and which part was not trained?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

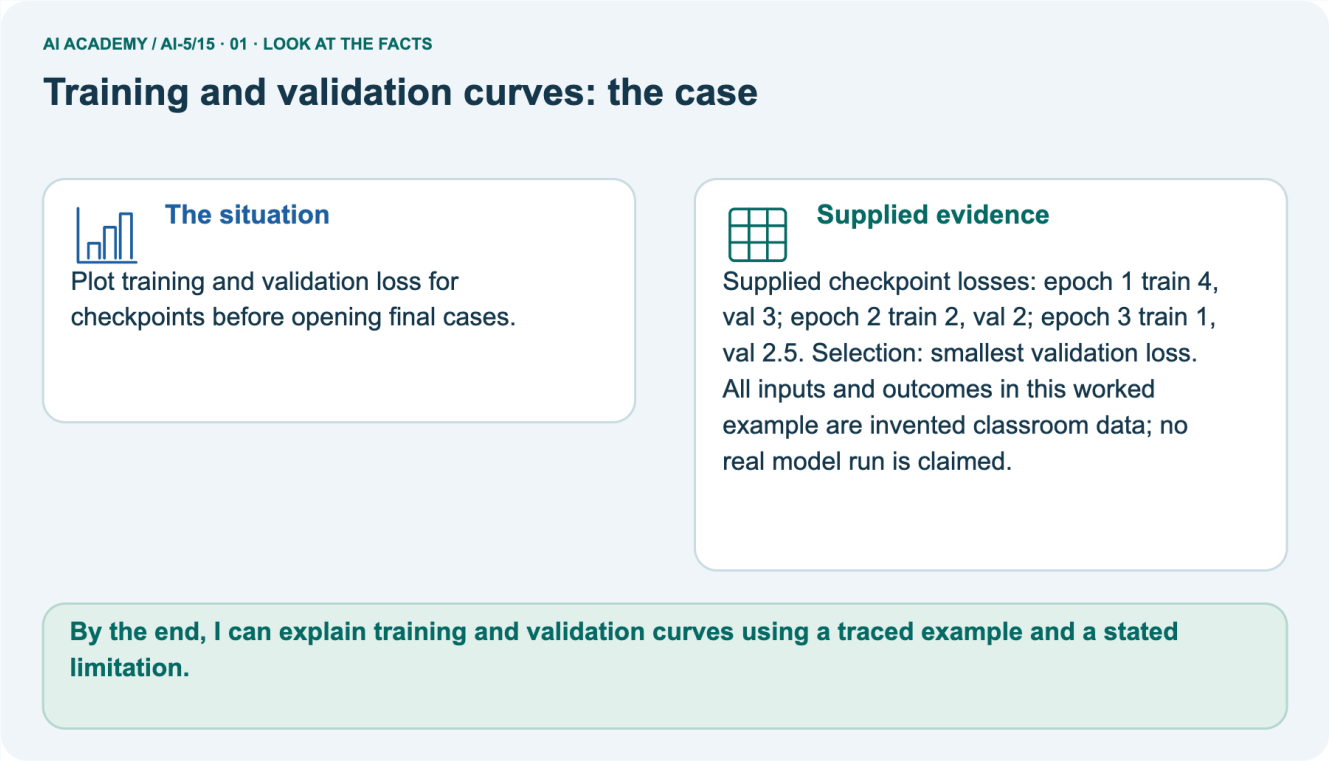
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Training and validation curves: the case



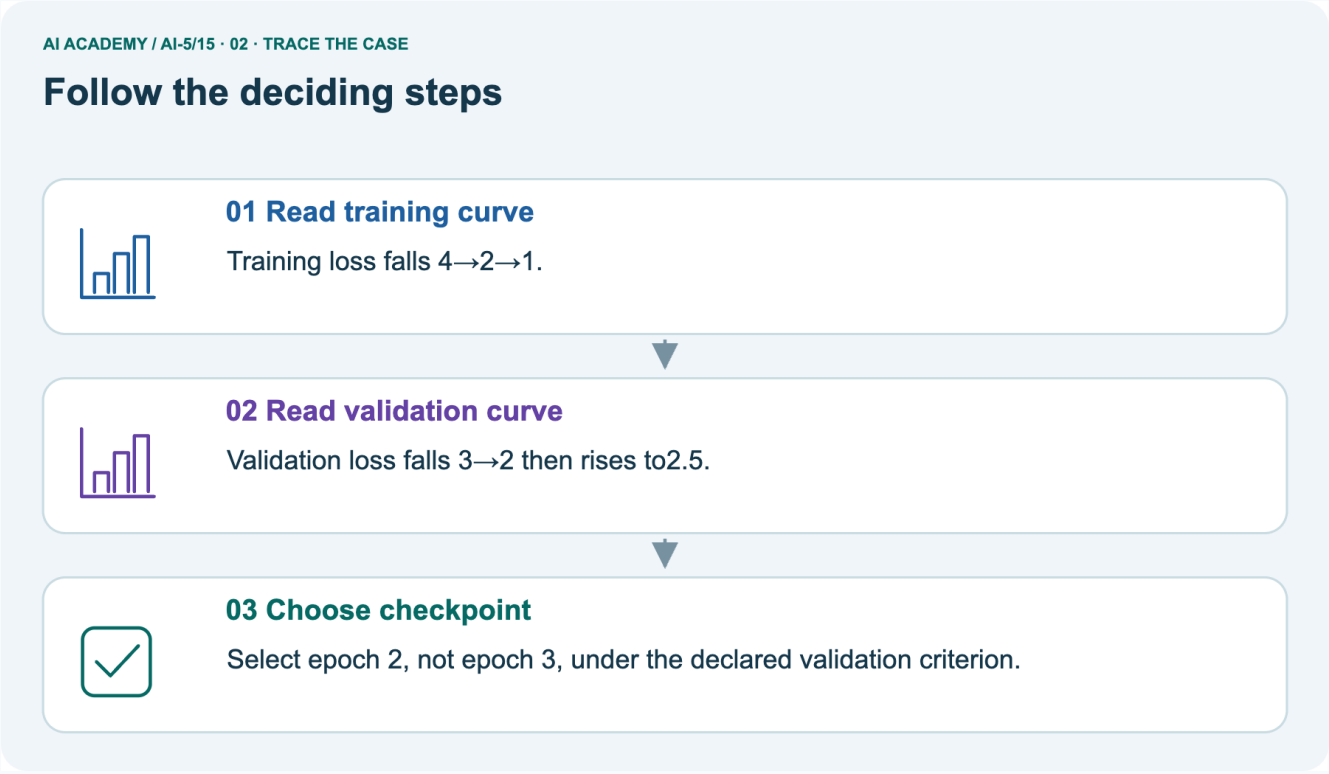
Follow the labelled picture with the matching explanation. Open the vector diagram

Plot **training** and validation loss for **checkpoint**s before opening final cases. Supplied facts: Supplied **checkpoint** losses: epoch 1 train 4, val 3; epoch 2 train 2, val 2; epoch 3 train 1, val 2.5. Selection: smallest validation loss.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read **training curve**: **Training** loss falls 4→2→1. → Read **validation curve**: Validation loss falls 3→2 then rises to2.5. → Choose **checkpoint**: Select epoch 2, not epoch 3, under the declared validation criterion.

Point to each numbered stage and explain which recorded **input** it uses.

03. Checkpoint losses

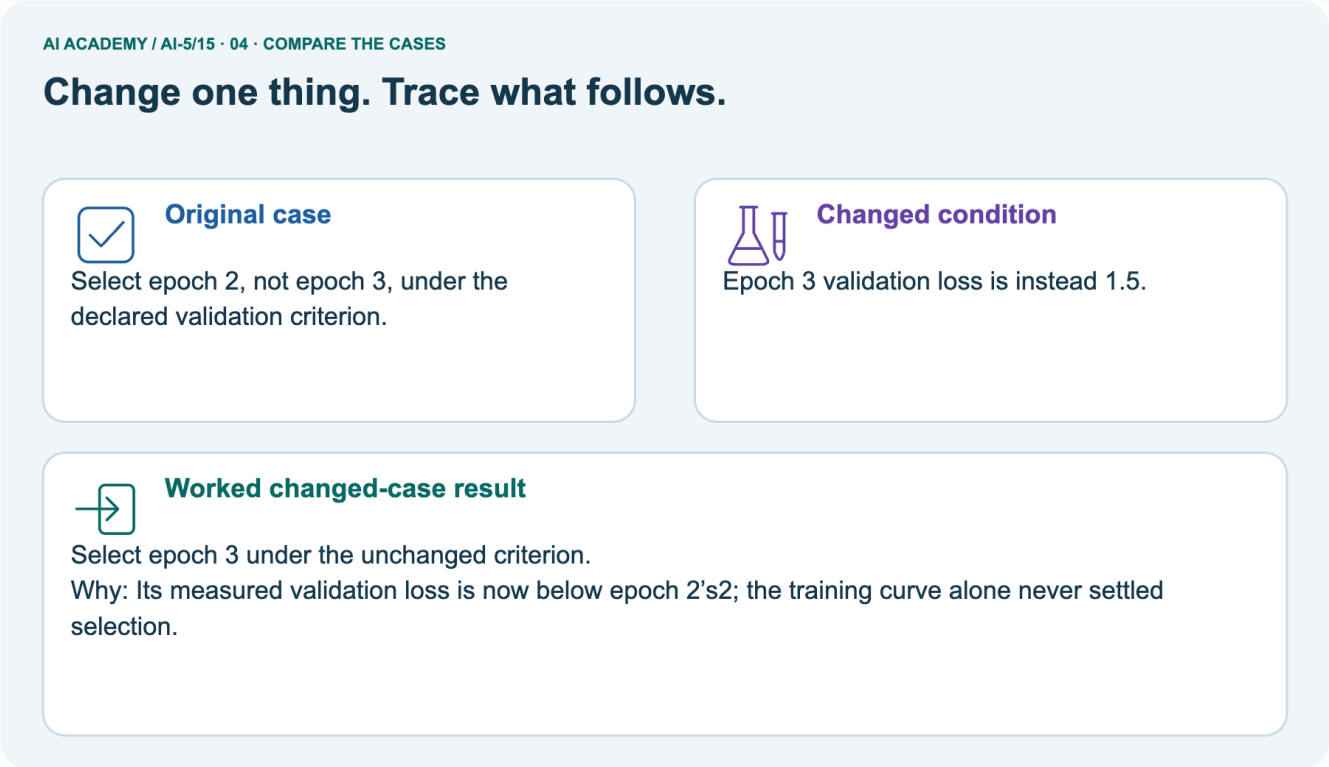


Follow the labelled picture with the matching explanation. Open the vector diagram

Checkpoint losses. epoch 1: **training** = 4, validation = 3. epoch 2: **training** = 2, validation = 2. epoch 3: **training** = 1, validation = 2.5.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



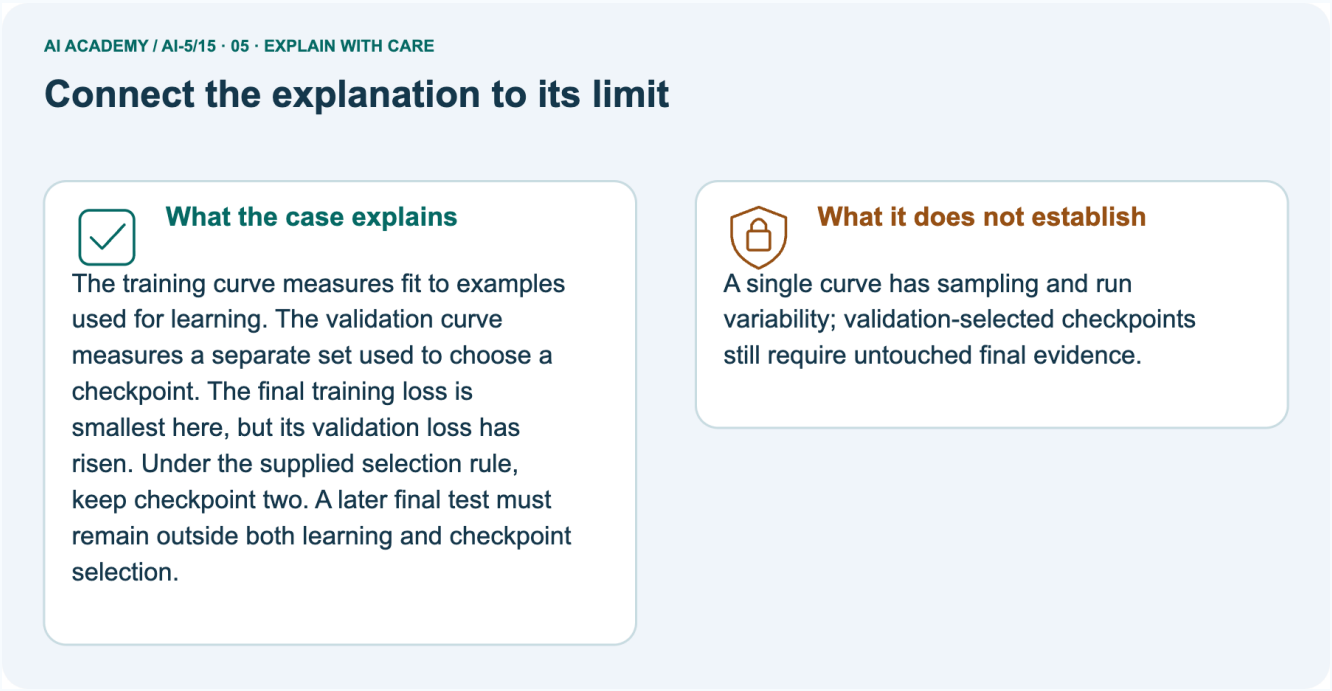
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Select epoch 2, not epoch 3, under the declared validation criterion. Changed condition: Epoch 3 validation loss is instead 1.5. Expected result: Select epoch 3 under the unchanged criterion.

Reason: Its measured validation loss is now below epoch 2's2; the **training curve** alone never settled selection.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

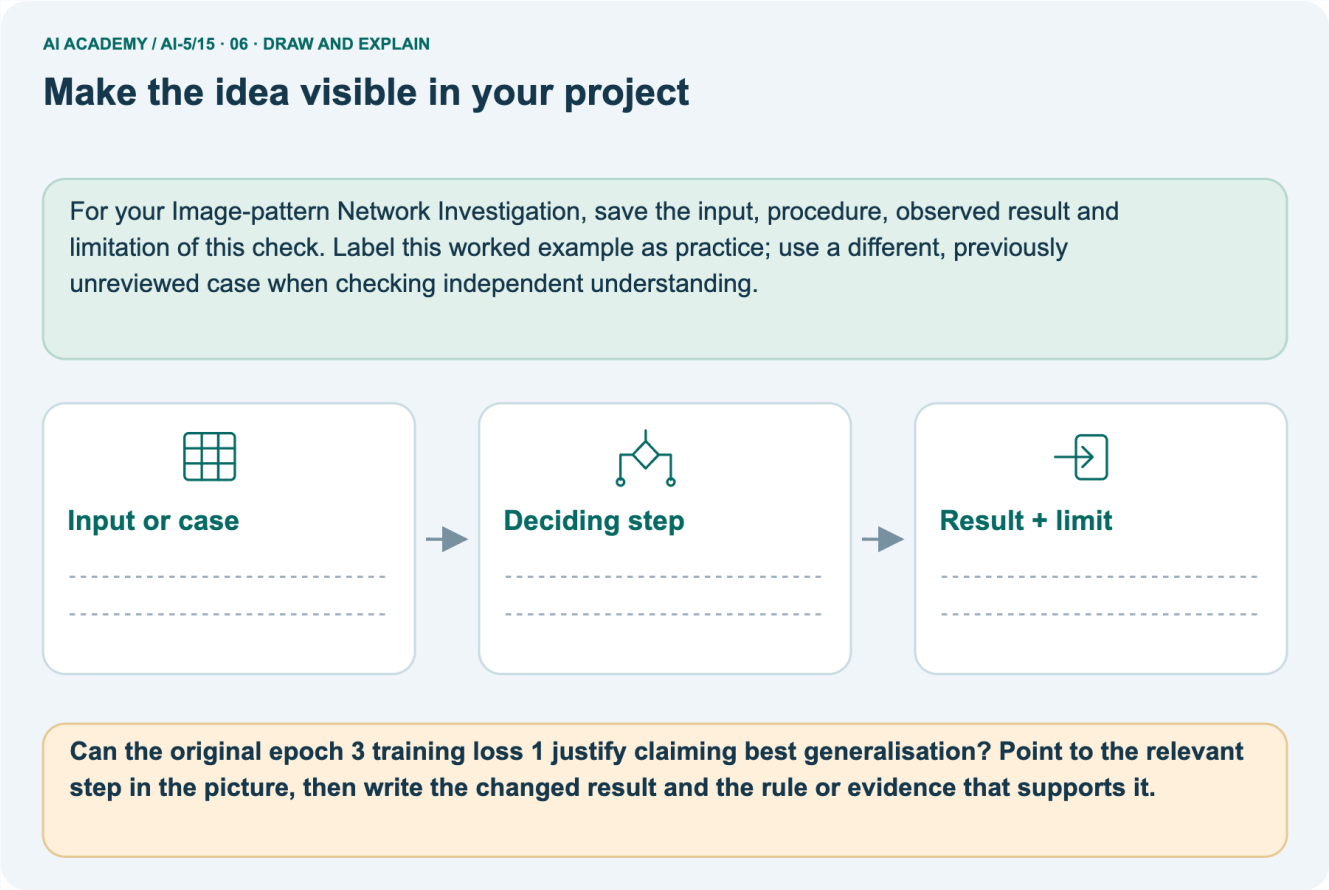
The **training curve** measures fit to examples used for learning. The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen.

Under the supplied **selection rule**, keep **checkpoint** two. A later final test must remain outside both learning and **checkpoint** selection.

Limit: A single curve has sampling and run variability; validation-selected **checkpoint**s still require untouched final **evidence**.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Training and validation curves: the case

02. Follow the deciding steps

03. Checkpoint losses

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.